

Homework discussion

December 2nd, 2025

Question 54

We suspect that a bridge player is cheating by putting an honor card (ace, king, queen, or jack) at the bottom of the deck when he shuffles so that this card will end up in his hand. In 16 times when he dealt, the last card dealt to him was an honor on 9 occasions. Are we confident that he is cheating?

Testing a probability model for chance situation.

Probability last card in the deck is an honor card:

Nick proposes $16/52$. Sounds good to me!

Probability model: 16 independent events “last card honor”

Question 54

9 of the 16 events occurred. What is the probability in the model that we get data like this or more extreme?

N = # of these 16 events that occur.

$N \sim$

X_i = indicator for honor card in last place in i th deal.

$X_i \sim \text{Bernoulli}(16/52)$

$N = X_1 + \dots + X_{16} \sim \text{binomial}(16, 16/52)$

Question 54

We now calculate (estimate) a p-value.

“Data like what we saw (9 of 16) or more extreme”

Calculate the probability of this event.

$P(N \geq 9)$

Why ($N \geq 9$)? Look at the expected number of honor cards $E[N] = 16(16/52) = 256/52$ about 5.

Durrett would like us to estimate using normal approximation (CLT).

Question 54

N is a sum of a fairly large number (16) of iid random variables. It's reasonable to think of N as approximately normally distributed.

$P(N \geq 9) = P(N \geq 8.5)$ approximately $P(\text{standard normal} \geq \text{z-score for } 8.5)$.

$\text{z-score for } 8.5 = (8.5 - E[N])/\sigma(N)$

Question 55

If both parents carry one dominant (A) and one recessive gene (a) for a trait then Mendelian inheritance predicts that 1/4 of the offspring will have both recessive genes (aa) and show the recessive trait. If among 96 offspring of Aa parents we find 30 are aa, is this consistent with Mendelian inheritance?

What is the probability model that we are trying to test?

We have 96 independent trials. And the outcomes of the trials are described by Mendelian inheritance. $P(\text{aa on trial } i) = 1/4$.

We observed 30 aa results. Is this plausible given the model?

Question 55

We need a “test statistic.”

N = # aa offspring.

Random variable in the probability model.

$N \sim \text{binomial}(96, 1/4)$

To test the model, we compute a p-value, i.e. a tail probability, for the observed data.

Is it $P(N \leq 30)$ or $P(N \geq 30)$?

$E[N] = 96(1/4) = 24$

Question 55

We need $P(N \geq 30)$ given $N \sim \text{binomial}(96, 1/4)$.

N is a sum of large number (96) of iid random variables, and so (CLT) it's reasonable to approximate N by a normal distribution.

$P(N \geq 29.5)$ is approximately $P(\text{standard normal} \geq \text{z-score for } 29.5)$

$\text{z-score for } 29.5 = (29.5 - 24) / \text{sigma}(N) = 5.5/4.24 = 1.30$

$\text{sigma}(N) = \sqrt{\text{Var}(N)}$

$\text{Var}(N) = 96(1/4)(3/4) = 24 (3/4) = 18$. $\text{sigma}(N) = \sqrt{18} = 3\sqrt{2} = 4.24$

Question 55

$P(\text{standard normal} \geq 1.30) = 1 - P(\text{standard normal} \leq 1.30) = 1 - 0.9032$
 $= 0.0968.$

p-value of 0.0968.

What do you make of that?

This p-value tells us that the probability of our model's giving data like what we saw or more extreme (for test statistic N) is around 10%. Rule of thumb is that you don't worry about your model unless the p-value is less than 5%.

Since the p-value is larger than 5%, we would certainly say that this experimental result is compatible with the Mendelian probability model for inheritance.